Exact Nearest-Neighbor Search with Composable Lower Bounds: Self-Configuring Hierarchical Pruning, In RAM and Out of Core

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ABSTRACT

Many applications cannot tolerate a missed true neighbor. Examples include deduplication, medical and legal retrieval, and scientific analysis. These applications need *exact* nearest-neighbor search. The standard tool for fast exact search is a cheap lower bound on each candidate's distance: if the bound already exceeds the best distance found so far, the candidate can be discarded without computing its full distance. This paper builds such bounds from truncated PCA projections. These bounds are tight when variance concentrates in a few dimensions. We further observe that they belong to a general family: any region that provably contains a set of points yields an exact lower bound, and different region shapes are tight for different data structures. We therefore compose two complementary families into a single exact cascade. The first is PCA truncation, which is tight for linearly correlated data. The second is a triangle-inequality ball bound around k-means centroids, which is tight for clustered data. The ball bound is exact for *any* choice of centers, so clustering quality affects only speed, never correctness.

Neither bound family helps on all data. The index therefore configures itself from a small sample. The cascade widths are chosen from the eigenvalue curve; fixed widths tuned for 128 dimensions misread 1536-dimensional OpenAI embeddings as unprunable, while widths placed at fixed variance targets recover them. A three-tier diagnostic then decides which levels to build: the eigenvalue curve and two cheap statistics nominate, and a measured trial of the real cascade decides. We report results in hardware-independent work units alongside wall-clock time, because wall-clock proved regime-dependent while work counts never moved. On four real datasets spanning 96 to 1536 dimensions (SIFT1M, GIST1M, DBpedia-OpenAI-1M, and DEEP10M), exact 1-NN search performs 3.9–20.8× less work than a full scan. The wall-clock gain grows with dimensionality, from 2.6× at 128 dimensions to 6.2× at 960. Exact k-NN generalizes the pruning radius to the k-th order statistic; it wins up to $k \approx 10$ and loses in RAM at $k = 100$. Out of core, the fetch set concentrates in few k-means clusters (27 of 1024 on the embeddings). A measured two-tier layout with `O_DIRECT` cold reads answers exact queries 7–17× faster than a full disk scan and reads 40–730× fewer bytes. We also compare against ADSampling, the state-of-the-art statistical pruner. In RAM the work counts are mixed. Out of core, however, ADSampling must touch every vector because of its 32-dimension floor, while our cascade fetches only 0.06–1% of the vectors. Exactness is verified on every query of every experiment. All code, diagnostics, and experiments are public.

PVLDB Reference Format:
Tai-Wu Chiang. Exact Nearest-Neighbor Search with Composable Lower Bounds: Self-Configuring Hierarchical Pruning, In RAM and Out of Core. PVLDB, 20(1): XXX-XXX, 2027.
doi:XX.XX/XXX.XX

PVLDB Artifact Availability:
The source code, data, and/or other artifacts have been made available at https://github.com/taiwuchiang-gmail/hierarchical-pca-search.

1 INTRODUCTION

Finding the exact nearest neighbor in a large, high-dimensional dataset is limited by the cost of computing full distances. Approximate methods such as LSH [7], HNSW [10], and IVF [8] trade recall for speed. However, some applications cannot accept a missed true neighbor. Examples include deduplication, medical and legal retrieval, and scientific analysis. For these applications, the alternative is exact pruning in the GEMINI tradition [4]: compute a cheap *lower bound* on each candidate's distance, and discard the candidate if the bound already exceeds the best distance found so far. Our starting point is such a bound from truncated PCA projections (§3.1). After the data is rotated onto its principal axes, the distance between k-dimensional prefixes is a lower bound on the true distance. A cascade of progressively finer prefixes, seeded by a dynamic Top-K probe, prunes over 99.9% of the SIFT1M dataset and returns provably exact results.[1]

That bound has a structural blind spot. PCA truncation is tight exactly when variance concentrates in a few principal directions, which means linearly correlated data. Clustered data can behave very differently. Its eigenvalue spectrum can be flat, which collapses the PCA cascade toward brute force. Yet the same data is highly compressible in another sense: points huddle near a modest number of centroids. No single bound family serves all data. Worse, a practitioner has no way to know, before building an index, which family (if any) their data supports.

This paper makes two moves. First, **exact bounds compose**. The maximum of valid lower bounds is itself a valid lower bound, so bound families with complementary blind spots can share one

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Proceedings of the VLDB Endowment, Vol. 20, No. 1 ISSN 2150-8097.
doi:XX.XX/XXX.XX

[1] Preliminary versions of this work appeared as two self-archived reports (doi:10.5281/zenodo.21387358, doi:10.5281/zenodo.22199955). This paper merges, extends, and supersedes them. The k-NN generalization, the eigen-curve width selection, three of the four datasets, the measured out-of-core results, and the ADSampling baseline are new.

cascade. We add a triangle-inequality *ball bound* around k-means centroids. It is the clustered-data complement of PCA truncation, and it is exact for *any* choice of centers: clustering quality affects only pruning power, never correctness. Second, **the composition is chosen by measurement**. A three-tier diagnostic runs on a sample in seconds. It rates the PCA levels from the eigenvalue decay, and it detects structure beyond the covariance with two cheap statistics. We show that statistics alone mispredict in both directions. The diagnostic therefore settles the question with a measured trial of the real cascade, and emits an ADD/SKIP verdict for each level.

There is also an economic motivation. An exhaustive flat scan requires the entire dataset to be resident in RAM, which is the most expensive byte in the machine. RAM costs roughly an order of magnitude more than NVMe per byte. The pruning cascade needs only its metadata resident: 8–40 bytes per vector, versus 512 bytes per vector for `fp32` data. At one billion vectors, this is 8–40 GB of RAM versus 512 GB. That is a commodity machine versus a specialty one, with exactness preserved.

Our contributions:

(1) **Exact hierarchical pruning from composable bounds.** A PCA-prefix cascade with a dynamic Top-K pruning radius, unified with the triangle-inequality ball bound under one containing-region principle. Each bound family covers the other's structural blind spot (§3).
(2) **A composed exact cascade, generalized to k-NN.** The cascade combines a ball level and the PCA levels, with a radius re-probe. Exact k-NN uses the k-th order statistic over evaluated points as the pruning radius. Storage overhead is about 8 bytes per point (§5).
(3) **Self-configuration that includes the cascade widths.** Widths are placed at fixed cumulative-variance targets on the eigenvalue curve; a fixed ladder is silently calibrated to one dimensionality and misreads others. The three-tier funnel then rules on each level, with verdicts derived from measured work ratios. Negative verdicts are included (§4).
(4) **Full-scale evidence on four real datasets** (96–1536 dimensions, 10^6–10^7 points), reported in hardware-independent work units. Wall-clock serves as cost-model validation. Negative controls and sample-to-full-scale checks are included (§6).
(5) **A measured out-of-core design.** We measure the fetch set's cluster locality, build a two-tier on-disk layout, and run cold-read benchmarks under `O_DIRECT`. Exact search is 7–17× faster than a full disk scan (§6.6).
(6) **An ADSampling baseline** in a shared work currency. It locates the honest boundary between statistical and deterministic pruning (§6.7).

2 BACKGROUND AND RELATED WORK

Lineage. Neither bound family is new, and we do not claim otherwise. Lower-bounding by dimensionality reduction descends from the GEMINI framework [4] and the VA-file [13]. Pruning by the triangle inequality around reference points is the basis of ball trees [11], vantage-point trees [14], GNAT [1], and Elkan's accelerated k-means [3]. The approximate cousins are FAISS IVF [8], which uses the same k-means geometry without exactness, and OPQ [6], a per-cell rotation that corresponds to the ellipsoid rung of the shape ladder in §3. Disk-resident ANN systems such as DiskANN [12] and SPANN [2] target the same memory-economics regime as we do, but they are approximate. The closest method in mechanism is ADSampling [5]. It also evaluates incremental prefix distances after a rotation, but it replaces the deterministic bound with a sequential hypothesis test under a *random* rotation. Its layout-level successor is PDX [9]. We measure against ADSampling directly in §6.7. Our contribution is threefold: the *exact composition* of the two bound families in one cascade, the *measured self-configuration* including the cascade widths themselves, and a diagnostic that also tells the practitioner when to walk away.

3 COMPOSABLE EXACT BOUNDS

3.1 The PCA Truncation Bound

Let $X, Y \in \mathbb{R}^d$ and let X', Y' be their images under the orthogonal rotation onto the dataset's principal axes. Rotation preserves the squared Euclidean distance, $L_2^2(X,Y) = L_2^2(X',Y') = \sum_{i=1}^{d}(x_i' - y_i')^2$. Let T_k retain the first k coordinates.

LEMMA 3.1. *For any truncation width $k < d$, $L_2^2\big(T_k(X'), T_k(Y')\big) \le L_2^2(X,Y)$.*

PROOF. The true squared distance is a sum of d non-negative terms; the truncated distance is the sum of the first k of the same terms, and $\sum_{i=k+1}^{d}(x_i' - y_i')^2 \ge 0$. □

If the prefix distance to a candidate already exceeds the current search radius r, the candidate is discarded without computing its full distance. This is the GEMINI no-false-dismissal condition [4]. Two choices turn the bound into an algorithm. The first choice is the rotation. Any orthogonal rotation satisfies Lemma 3.1. PCA makes the bound *tight*: by the Karhunen–Loève theorem, it packs the maximum variance into the first k coordinates, and therefore, on average, the largest share of any pairwise distance (quantified in Eq. (3)). In other words, exactness is unconditional but quality is not. This separation recurs throughout the paper. The second choice is the source of the radius. Instead of a static threshold, a Top-K probe at the coarsest width computes exact distances for the $K = 50$ most promising candidates and sets r to their minimum. The prefix distance is a lower bound, so this radius is safe from the first comparison on. No prior knowledge of the neighbor distance is needed. A cascade of progressively finer prefixes then re-prunes the survivors at each width. On SIFT1M, widths $(8, 16, 32, 64)$ discard all but an average of 564 of 10^6 vectors before any full-dimensional evaluation. Section 4 replaces these fixed widths with widths derived from the data's own eigenvalue curve.

3.2 The Containing-Region Principle

Let $R \subseteq \mathbb{R}^d$ be any region that provably contains a set of points C. Then for every $p \in C$ and any query q,

$$\|q - p\| \;\ge\; \mathrm{dist}(q, R), \tag{1}$$

so $\mathrm{dist}(q,R)$ is an exact pruning bound for every member of C at once. A bound family is therefore a *shape* family, and a shape is admissible when (a) containment is guaranteed and (b) $\mathrm{dist}(q,R)$ is

Table 1: Shape families and their exact bounds. Each is tight for a different data structure; each mirrors a quantizer family.

Shape	Bound cost	Tight for	Quantizer twin
ball	$\lvert \lVert q-c \rVert - d_p \rvert$	clusters	VQ
ellipsoid	per-axis scaled	stretched clusters	OPQ
slab	projection residual	local low rank	local PCA
cone×annulus	closed form	radial/gain data	gain–shape

cheap. Table 1 lists the natural ladder. Each geometric family is the twin of a quantizer family: a quantizer's codeword-plus-distortion model *is* a containing region. We use the first two rungs; the others are available under the same principle.

3.3 The Ball Bound

Cluster the dataset with k-means. For each point p assigned to centroid c, store the scalar $d_p = \|p - c\|$ once, offline. At query time, compute the k centroid distances $\|q - c_j\|$ (with $k \ll N$), and the triangle inequality gives, for every point,

$$\|q - p\| \;\geq\; \big|\, \|q - c_{a(p)}\| - d_p \,\big|, \tag{2}$$

where $a(p)$ is p's cluster. Evaluating (2) for all N points requires one gather and one subtraction per point—*no distance computations.* A coarser, per-cluster form uses the cluster radius $R_j = \max_{p \in j} d_p$: if $\|q - c_j\| - R_j$ exceeds the current radius, the entire cluster is eliminated in one comparison, which in an out-of-core layout skips a whole contiguous block (§5.4).

3.4 Exactness Is Unconditional; Quality Is Not

Inequality (2) is the triangle inequality, so it holds for *any* reference points. k-means enters only because it minimizes the mean d_p^2, which makes the bound *tight.* Three corollaries follow. First, stale centers remain correct as the dataset drifts. Pruning degrades gracefully, and re-clustering can wait until a diagnostic calls for it. Second, insertion is $O(1)$: assign the new point to its nearest existing centroid and store d_p. No re-clustering is required for correctness. Third, the composed index mirrors the PCA bound's property from §3.1. There, any orthogonal rotation is exact and PCA makes it tight. Here, any centers are exact and k-means makes them tight. The maximum of lower bounds is a lower bound, so the two families compose freely in one cascade.

A note on terminology. Throughout this paper, each pruning step of the cascade is called a *level.* The *ball level* applies the ball bound (2) to all points. Each *PCA level* applies the prefix bound of Lemma 3.1 at one width. A query passes through the levels in order, so we speak of a level's *survivors*: the candidates that its bound could not discard.

3.5 Complementarity

The PCA truncation bound is tight if and only if the eigenvalue spectrum decays steeply. This is a *global, linear* property (Figure 1, left). The ball bound is tight if and only if points sit close to their centroids relative to the spread of $\|q - c\|$. This is a *local* property. It holds for multimodal data, and more generally for data of low

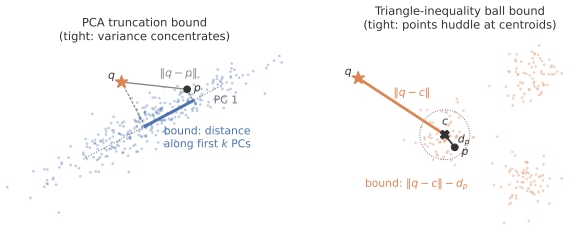


Figure 1: The two exact bound geometries. Left: the PCA truncation bound is the distance between the projections onto the first k principal components—tight when variance concentrates along few directions. Right: the ball bound $\|q - c\| - d_p$ from the triangle inequality—tight when points huddle near their centroids. Each is loose exactly where the other is tight.

intrinsic dimension, where d_p and the centroid distances spread widely instead of concentrating. The two blind spots are disjoint by construction. Our clearest demonstration is a synthetic dataset of 512 tight clusters whose centers span all dimensions. Its spectrum is nearly flat: 95% of the variance needs 111 of 128 dimensions, and the PCA cascade manages only 2.1×. Yet the ball bound prunes this dataset to 3% before a single projection is computed.

4 THE SELF-CONFIGURATION FUNNEL

The funnel answers, from a sample and in seconds, "which levels should this index have for *my* data?" Figure 2 shows its shape: two statistical tiers that may only *nominate*, and one measured tier that alone *decides.*

Tier 1 — eigenvalue decay, which also *picks* the widths. Fit PCA on a sample. The eigenvalue curve is not merely a heuristic signal: in the PCA basis the coordinates of the difference between two independently drawn points are uncorrelated with variances $2\lambda_i$, so for typical candidate pairs

$$\frac{\mathbb{E}\, L_2^2\big(T_k(X'), T_k(Y')\big)}{\mathbb{E}\, L_2^2(X, Y)} \;=\; \frac{\sum_{i \leq k} \lambda_i}{\sum_i \lambda_i}, \tag{3}$$

which is the cumulative explained variance at k. The curve therefore reads directly as the *expected fractional tightness* of the width-k bound on the population the cascade prunes. In this sense, each dataset's curve is its prunability signature (Figure 3). Two blind spots follow from the same statement, and each one motivates a later tier. First, Equation (3) is an expectation over pairs, but pruning is a tail event: a candidate is discarded only when its prefix distance clears the radius. Scalar summaries of the curve can therefore mislead. For example, GIST1M needs 303 of 960 dimensions for 95% variance, which looks like a flat spectrum by the usual standard. However, its steep *head* (48% by eight components) is what the early levels actually use, and the measured verdict is STRONG. Second, the spectrum captures only second-order structure. Clustered structure that no rotation can expose is invisible to it. Tier 2 screens for that structure, and tier 3 settles the question.

Preliminary versions of this work used a fixed ladder $(8, 16, 32, 64)$. We discovered that this ladder is silently calibrated to $d \approx 128$.

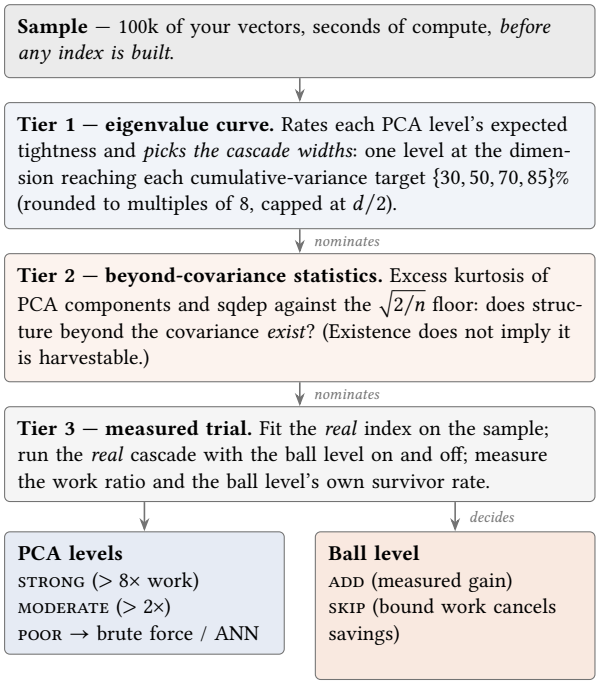


Figure 2: The self-configuration funnel. Tiers 1–2 are cheap statistics, and they may only *nominate*. Tier 1 also derives the cascade widths from the eigenvalue curve. Tier 3 is a measured trial of the real cascade, and it alone *decides*. Section 6.5 shows that the sample verdicts hold at 10^6–10^7 scale. The tier-2 mispredictions in the text (SIFT and latent-8, in opposite directions) are why no statistic can replace the trial.

On 1536-dimensional OpenAI embeddings, it stops at 4% of the dimensions and prunes almost nothing: 68% of candidates survive the 64-D level. As a result, the funnel misreads clearly prunable data as hopeless. The fix makes tier 1 constructive. We place one level at the dimension reaching each cumulative-variance target $\{30, 50, 70, 85\}$%, rounded up to a multiple of 8 and capped at $d/2$. We measured this rule against hand-tuned ladders on 100k samples, using the work ratio. The rule ties on SIFT (8.2× vs. 8.5×). It *wins* on GIST1M (15.2× vs. 9.3×) and on DBpedia-OpenAI (8.2× vs. 7.6×). A self-configuring index cannot have a hard-coded ladder. The eigen curve nominates the widths, and the trial below still decides.

Verdicts are work ratios, not survivor fractions. We caught a second dimension-calibration bug in the same way. Rating prunability by the final survivor fraction implicitly assumes one dimensionality. GIST1M's 6.9% finalists scored "poor" under a threshold tuned for SIFT. However, at $d = 960$ the same 6.9% means 9.3× fewer terms, because a scan always pays $n \cdot d$. The verdict now derives from the measured work ratio (sample terms vs. $n \cdot d$), which is dimension-aware by construction.

Tier 2 — beyond-covariance statistics. Tier 2 computes two cheap statistics on the PCA-rotated sample. The first is the mean excess kurtosis of the components, which measures marginal non-Gaussianity. The second is sqdep, the mean $|\mathrm{corr}(z_i^2, z_j^2)|$ over component pairs. It measures joint dependence that no rotation can

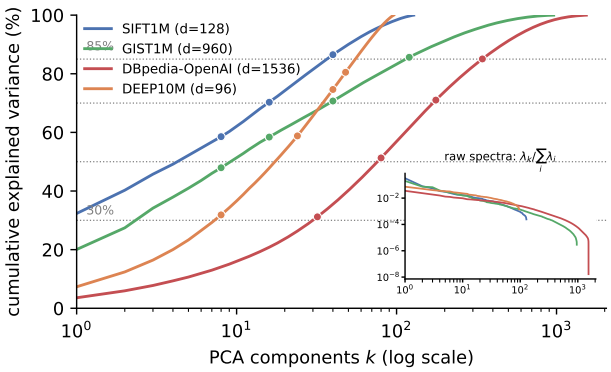


Figure 3: Cumulative explained variance on 100k samples, with components on a log scale. Each curve is a dataset's prunability signature: by Eq. (3), the curve equals the expected fractional tightness of a width-k bound. Dots mark the widths the funnel selects at the $\{30, 50, 70, 85\}$% targets (dotted lines). SIFT and GIST have steep heads. DBpedia's curve is shifted to the right, which is why a fixed d=128 ladder misreads it as unprunable. DEEP is the flattest, which matches its MODERATE verdict; its last width sits below the 85% line because of the $d/2$ cap. The inset shows the raw spectra (per-component variance fraction, log–log), for readers who want the decay rates directly.

remove, with a finite-sample noise floor of about $\sqrt{2/n}$. These statistics detect that structure beyond the covariance *exists*. They do not tell whether it is *harvestable*.

Tier 3 — the measured trial. Tier 3 fits the real hybrid index on the sample and runs the real cascade twice, with the ball level on and off. It reports the measured ratio and the ball level's own pruning rate. The verdict is ADD or SKIP.

Tier 3 must be the decider. The evidence is that the statistics mispredict in *both* directions. On SIFT, tier 2 fires: sqdep = 0.032 is 7× the noise floor, so real beyond-covariance structure exists. Yet the trial says SKIP (1.0×), because the ball bounds prune only 50–59% and the bound work cancels the savings. On a latent-dimension-8 Gaussian, tier 2 is clean: kurtosis is near 0 and sqdep sits at the floor. Yet the trial says ADD (1.9×), because low intrinsic dimension spreads d_p widely and the bounds prune to 23% with zero multimodality. We call the ball level's measured value its *gain*: the wall-clock ratio of the cascade with the ball level to the cascade without it. Across all datasets, this gain is a monotone function of the ball level's own survivor rate (Figure 4). Survivors of 3% give a gain of 4–14×; 15–23% give 2–3×; 50–59% give parity; and 100% gives 0.8×, a measured loss. The trial measures exactly this quantity. No statistic substitutes for it.

5 THE HYBRID ALGORITHM

5.1 Offline

The offline build has three steps. First, fit PCA; the covariance is accumulated in chunks, so no full `float64` copy of the dataset is ever materialized. Second, select the cascade widths from the eigen

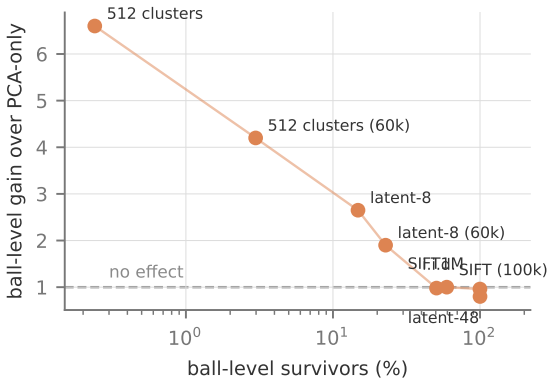


Figure 4: The measured ball-value law across all eight datasets and trials (parity code): the ball level's wall-clock gain over the PCA-only cascade is a monotone function of its own pruning rate. The funnel's tier-3 trial measures precisely the x-axis quantity, which is why it predicts full-scale outcomes from a sample—including the negative verdicts below the dashed line.

curve (§4). Third, fit k-means with $k = \text{clip}(\sqrt{N}, 64, 4096)$, using MiniBatch (24 s on SIFT1M for k=1024). For each point we store a cluster id (`int32`) and d_p (`float32`), 8 bytes in total, plus the $k \times d$ centroids and the per-cluster radii. The choice of k is measured, not tuned. A sweep over $k \in \{8, 64, 256, 1024, 4096\}$ on SIFT1M shows ball-level survivors falling monotonically from 81% to 43%, while fit cost rises from 4 s to 47 s. $\sqrt{N}$ sits at the knee of this gain-versus-cost curve, so it is the default.

5.2 Query Pipeline

(1) k centroid distances ($k \cdot d$ operations).
(2) Ball bounds (2) for all N points: one gather and one absolute difference each.
(3) **Probe**: exact distances to the PROBE= 50 points with the smallest ball bounds seed a safe initial radius r (the Top-50 probe of §3.1, seeded by ball bounds instead of coarse projections).
(4) Ball pruning: retain points with bound$^2 < r^2$.
(5) PCA cascade (8, 16, 32, 64) on the survivors, with a **re-probe** at the first PCA level: before pruning, exact distances to the 50 best 8-dimensional candidates tighten r. Without this step the ball-seeded radius is loose and the cascade degrades (8-D survivors 27.2% versus 14.6% with it); with it, the cascade matches PCA-only tightness level-for-level.
(6) Full-distance check on the final survivors. The result is provably identical to brute force, and is additionally verified against the official ground truth in every experiment.

5.3 Exact k-NN

For k-NN, the pruning radius generalizes from the best evaluated distance to the k-th smallest exact distance among the points evaluated so far. The k-th order statistic over a *subset* can only overestimate the true one. Therefore, any point whose lower bound reaches this radius provably cannot enter the k-NN set. One subtlety carries an exactness bug if it is missed. The evaluated multiset can contain a point twice, because the probe and re-probe sets overlap. The k-th order statistic of a multiset with duplicates is biased low, and a radius that is too small *over*-prunes. The minimum (the 1-NN special case) is not affected by duplicates, but the k-th order statistic is. The radius must therefore be taken over *distinct* evaluated points. The probe size also grows to $\max(50, k)$, so that the initial radius is a valid k-th statistic. Correctness is verified against brute force across datasets, both cascade paths, and k up to $n - 1$. A second subtlety surfaced during verification. Integer-valued descriptors make exact distance *ties* real: SIFT query 13 has two points at squared distance exactly 61334, and only one of them is in the official ground truth. k-NN sets are therefore unique only up to ties at the boundary.

5.4 Out-of-Core Layout

Pruning fragments I/O only if the layout ignores it. This is the classic index-versus-scan crossover: hundreds of scattered small reads can lose to one full sequential scan. Our two-tier design keeps *resident* only what pruning decisions touch. The resident tier holds the rotated sketch of the first max(levels) PCA dimensions, the ball metadata (`assign`+d_p, 8 bytes per point), the centroids, and the per-cluster offsets. Together this is 2–8× smaller than the data. Full-dimensional vectors live on disk in two interchangeable files. The first keeps the original order, so each fetch is one small aligned read. The second is *cluster-major*: each k-means cluster is a contiguous, page-aligned block (under 0.4% padding), written sequentially into `fallocate`-preallocated extents. The entire cascade (probe, re-probe, and all pruning levels) runs on the resident tier. The disk is touched only by the fetch set: probe, re-probe, and finalists. Section 6.6 measures exactly which points that is.

Two structural facts make the layout matter. First, the cluster-level bound alone is a weak block-skipper: on SIFT1M, 40–52% of clusters survive it. Second, the fetch set is far more concentrated than uniform. The nearest neighbors of a query are near each other, so they share clusters, and the final fetch becomes a few large reads. This is the wisdom of the VA-file [13], SPANN [2], and DiskANN [12]: scan small sketches sequentially, and fetch survivors in a layout-aware way. The same k-means that supplies the bound also supplies the layout. Section 6.6 shows that it earns its keep as a *partitioner* even on datasets where the funnel rejects it as a *pruning level*.

6 EVALUATION

Metrics policy. Primary claims are stated in hardware-independent work units. The reason is that wall-clock proved regime-dependent: the same algorithm measured 4.7×, 3.6×, 2.7×, and parity under changes of numeric type, machine load, and implementation tuning, while its work counts never moved (details below). The primary metrics are: (1) full-vector fetches per query, which is the

Table 2: SIFT1M, 100 official queries, medians of 5 repetitions, implementation parity. All methods return the ground-truth neighbor on every query.

Method	ms/query	Speedup	Survivors per level (%)
brute (NumPy)	167.3	1.0×	—
PCA-only	61.1	2.7×	14.7 / 5.2 / 0.8 / 0.06
hybrid	62.2	2.7×	ball 50.6 → 14.6 / 5.1 / 0.7 / 0.05
FAISS-flat, 1 thread	29.3	5.7×	(scans 100%)
FAISS-flat, batch-100	5.9	28×	(scans 100%)

out-of-core I/O currency; (2) total bytes touched per query, *including the cost of the pruning machinery itself*; (3) resident RAM bytes per vector; and (4) the survivor cascade, which is the algorithmic fingerprint. Wall-clock appears as validation of a first-order cost model ($t \approx$ bytes/bandwidth + ops/throughput), with FAISS supplying the tuned-constants reference. A reader can therefore plug their own machine's constants into our work numbers and predict their own outcome.

Setup. Intel i9-9900K (8C/16T, AVX2+FMA, no AVX-512), dual-channel DDR4, 30 GB usable RAM; NVMe SSD measured under `O_DIRECT` at 2.04 GiB/s sequential (8 MiB reads) and 80 µs per random 4 KiB read; `faiss-cpu` 1.15 dispatching its AVX2 build; NumPy and OpenBLAS at the AVX2 tier—baseline and FAISS use the same instruction set. SIFT1M in-RAM timings (Table 2) are medians of 5 repetitions on a quiet machine with warm cache; the multi-dataset, k-NN, and out-of-core tables are medians or means over 100 queries in a single repetition on a quiet machine (work counts and byte counts are exact and run-invariant; only the millisecond columns carry run noise). Exactness is verified on every query of every experiment—against official ground truth where one exists, against in-harness brute force otherwise. Results are CPU-governor-insensitive (medians within 2%).

6.1 SIFT1M

Table 2 gives the in-RAM picture, and Figure 5 shows the cascade behind it. The PCA cascade alone reaches 2.7×. **The ball level is a wall-clock tie on SIFT**: its 50% pruning saves about as much as its bound computation costs. This is exactly what the sample trial predicts (SKIP, §4). On SIFT, the hybrid's value is therefore limited to the out-of-core and residency settings (§5.4). The wall-clock case for the ball level rests on clustered and low-intrinsic-dimension data (§6.4). The survivor cascade exactly reproduces the preliminary reports'.

Table 3 states the same experiment in the primary currency. Three facts that the wall-clock hid now become explicit. First, the cascade's *algorithmic* reduction is 10.2×. This number is machine-free, and it was invariant across every run in this paper, while the millisecond columns wobbled with load. Second, FAISS-flat performs 10–15× more work than the cascade and still wins the in-RAM wall-clock. The separation between engineering and algorithm is now quantitative rather than rhetorical. Third, the hybrid cuts terms further (12.6 → 8.7 M) but adds 1.0 M bound operations of a different kind: gathers rather than streaming reads. This is why its wall-clock ties PCA-only despite the smaller term count.

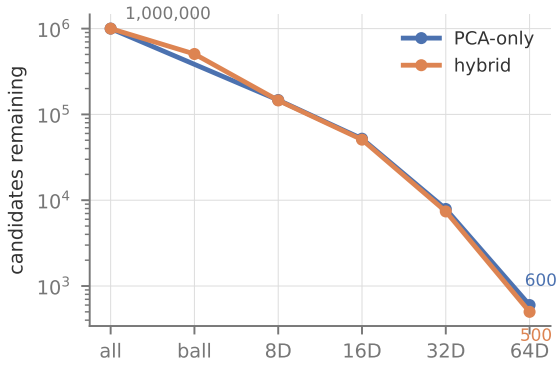


Figure 5: Candidates remaining after each level on SIFT1M (log scale, average over 100 official queries). The cascade discards four orders of magnitude before any full-distance evaluation; the hybrid enters the PCA levels with half the population and finishes equally tight. This profile is the algorithmic fingerprint: it was invariant under every change of numeric type, machine load, and implementation tuning in this paper.

Table 3: Hardware-independent work per query on SIFT1M. Terms are per-dimension $(x-y)^2$ evaluations; bound ops are per-point ball-bound evaluations, charged separately. Counts derive analytically from the survivor statistics (`count_ops` in the repository)—an exhaustive scan is $N \cdot d$ terms by definition, so no instrumentation of compiled libraries is needed. These columns are identical on every machine and every run.

Method	Terms (M)	Bound ops (M)	Terms vs brute
brute / FAISS-flat	128.0	—	1.0×
PCA-only	12.60	0	**10.2**×
hybrid	8.67	1.00	14.8×

It is also why work accounting that omits the pruning machinery would mislead.

FAISS `IndexFlatL2` wins the in-RAM race outright. It scans all 10^6 vectors in 29.3 ms on a single thread. The gap is fusion, not SIMD. Both sides run AVX2, but FAISS's kernel makes one pass over the data (~512 MB of traffic per query), while NumPy materializes temporaries (~3 passes, 1.5–2 GB). That is a 5.6× advantage at identical work. This comparison scopes our claim rather than defeating it. The cascade touches 1,700× fewer full vectors. It needs only 8–40 resident bytes per vector, against 512 for any flat scan. In addition, its bounds compose with FAISS-grade kernels rather than compete with them: the cascade could drive such a kernel over its survivor sets. Disk-resident ANN systems occupy the same economics but give up exactness. Table 4 summarizes the resulting trade among speed, work, and the machine's most expensive resource.

Table 4: The exactness–speed–work–memory trade-off on SIFT1M (fp32, $10^6 \times 128$). Flat scans must hold every vector resident (512 B/vector). The cascades keep only pruning metadata resident and page full vectors from storage (§5.4): the 8-dimensional sketch costs 32 B/vector, and the ball metadata costs 8 B/vector. Parentheses give the RAM saving relative to a flat scan. The hybrid's work is written as distance terms plus ball bound-ops (as in Table 3); the two are kept separate because a bound-op is cheaper than a term, so they are not summed. The last row situates disk-resident ANN systems. Their residency (taken from their own reports, not measured here) is comparable to the cascades', but they answer the approximate problem. This work occupies the exact-at-cascade-residency quadrant.

Method	Exact	ms/query	Terms (M)	Resident RAM
FAISS-flat, batch-100	yes	5.9	128.0	512 MB (1×)
FAISS-flat, 1 thread	yes	29.3	128.0	512 MB (1×)
brute (NumPy)	yes	167.3	128.0	512 MB (1×)
PCA-only	yes	61.1	12.6	32 MB (16×)
hybrid	yes	62.2	8.7 +1.0	8–40 MB (13–64×)
IVF / disk ANN [2, 8, 12]	no	—	—	∼8–64 MB

Table 5: Exact 1-NN at full scale, eigen-curve-selected levels, PCA-only cascade. Terms count per-dimension $(x-y)^2$ evaluations; a scan is $n \cdot d$ by definition. All queries exact.

Dataset	n	d	Wall	Terms (M)	Work red.
SIFT1M	10^6	128	2.6×	12.6	10.2×
GIST1M	10^6	960	**6.2**×	46.1	**20.8**×
DBpedia-OpenAI	10^6	1536	6.0×	119.6	12.8×
DEEP10M	10^7	96	1.2×	244.0	3.9×

Why wall-clock is the wrong headline (measured). Two confounds were caught only by asking why ratios changed between runs. The first is numeric type. sklearn's PCA silently outputs `float64`, and the bandwidth-bound brute baseline is 1.86× more expensive in `float64`. A 2 × 2 decomposition on identical code measures the same cascade at 3.6× in f64 and 2.7× in f32. The f64 number matches the first preliminary report's regime and its reported 4.3×. The second confound is implementation parity. An unnecessary full-array gather in our own PCA-only path inflated the ball level's apparent contribution to 2.0×. Fixing it revealed the tie in Table 2. Machine load adds a further multiplier: background bandwidth contention slows the scan-bound baseline the most, which flatters all pruners. Survivor counts were invariant through every one of these changes. This is why we adopted the metrics policy above.

6.2 Four Real Datasets: the Gain Grows with Dimension

Table 5 answers a question that SIFT alone cannot: does anything beyond SIFT prune? Everything we measured does. Moreover, the wall-clock gain *grows* with dimensionality: 2.6× at 128 dimensions, 6.2× at 960, and 6.0× at 1536. The reason is simple. Brute force pays the full dimension for every point, while the cascade pays only for the intrinsic structure. The most important result concerns the modern learned embeddings (DBpedia abstracts under OpenAI `ada-002`). Their flat spectra (664 of 1536 dimensions for 95% variance) looked like the natural failure mode of our approach. In fact, they prune to 0.64% finalists once the cascade widths respect the eigen curve (§4). With the *fixed* (8, 16, 32, 64) ladder, the same data measures only 1.3× and reads as unprunable. DEEP10M is the honest boundary case. Its 96 dimensions are already-compressed deep features, and they leave a 3.9× work reduction but only 1.2× in-RAM wall-clock. The funnel's sample verdict (MODERATE, "useful out-of-core") says exactly this in advance, and §6.6 confirms where its value lives. Finally, on all four datasets, the ball level's sample verdict (SKIP everywhere; survivor rates 50–99%) was confirmed at full scale. The ball-value law of §4 holds on real data at 10^6–10^7 points.

Table 6: Exact k-NN, PCA-only cascade, wall-clock speedup over brute force and (in parentheses) work reduction in terms. All queries exact; k=100 additionally matches official ground truth up to boundary ties.

Dataset	$k = 1$	$k = 10$	$k = 100$
SIFT1M	2.6× (10.2×)	1.9× (6.2×)	0.7× (1.6×)
GIST1M	6.2× (20.8×)	4.0× (10.3×)	0.9× (1.5×)
DBpedia-OpenAI	6.0× (12.8×)	3.1× (5.8×)	0.8× (1.2×)
DEEP10M	1.2× (3.9×)	0.8× (2.0×)	0.3× (0.6×)

Table 7: Structure regimes (synthetic, $n = 200$k, $d = 64$ for the first two; $n = 60$k, $d = 128$ for the funnel rows), implementation parity. "Ball survivors" is the ball level's own pruning rate—the quantity the funnel's trial measures.

Dataset	PCA-only	Hybrid	Ball surv.	Verdict
512 tight clusters	2.1×	**13.9**×	3%	ADD
latent-8 Gaussian	2.1×	5.6×	15–23%	ADD
SIFT1M	2.7×	2.7×	50–59%	SKIP
latent-48 Gaussian	1.9×	0.8× of PCA	100%	SKIP
i.i.d. Gaussian	0.3×	—	100%	SKIP (all)

6.3 Exact k-NN

Table 6 reports the k-NN generalization (§5.3) with deliberate honesty. The method wins up to $k \approx 10$, and it *loses* the in-RAM wall-clock at $k = 100$ on every dataset. The mechanism is structural, not incidental. The pruning radius is the k-th best distance, which grows with k, and the survivor counts grow with it. At k=100 the ball level passes 99–100% of the points, and the cascade approaches a scan with extra overhead. However, the k=100 work reductions (1.2–1.6× on the million-scale sets) still translate into fewer bytes fetched out of core. That is where large-k exact queries should run (§6.6).

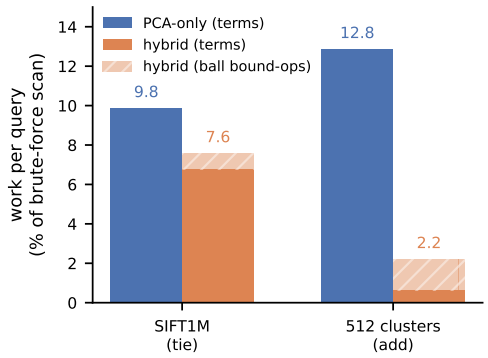


Figure 6: Per-query work of the two cascades, as a percentage of a brute-force scan, on a tie dataset (SIFT1M) and an add dataset (512 clusters). Bars are terms; the hatched segment is the ball level's bound-ops, charged separately. On SIFT the bars are nearly equal, which is the wall-clock tie. On the clustered data the hybrid does far less work. Both cascades return the same exact result and reach the same survivors (Table 7); only the work differs, which a survivor-count plot cannot display.

6.4 Structure Regimes

Table 7 spans the design space, and it deliberately includes the negative results. The flat-spectrum clustered dataset is PCA's blind spot. There, the ball level prunes to 3% before any projection is computed, and the hybrid reaches 13.9×. Its bound requires no multiplications, so it undercuts even the 8-dimensional scan. Figure 6 makes the contrast with SIFT concrete in the work currency. On SIFT the two cascades do almost the same work (9.8% vs 7.6% of a scan), which is the tie. On the clustered data the hybrid does far less work (2.2% vs 12.8%), and most of its small cost is the ball bound-ops, not distance terms. This is what a survivor-count plot cannot show: both cascades reach the same survivors, but the hybrid reaches them with much less work. On the latent-48 blob, the ball level is measured to *hurt* (0.8×): 100% of points survive, so the level is pure overhead. On i.i.d. noise, the diagnostic correctly reports that nothing works and that brute force or ANN should be used. The i.i.d. row also shows the PCA cascade itself losing to brute force (0.3×). That is the honest boundary of the entire approach.

6.5 Extrapolation

Does the sample predict the full dataset? On SIFT, the funnel's trial on a 100k sample returns 1.0× (SKIP), and the full-1M parity benchmark measures exactly that (61.1 versus 62.2 ms). The check extends to every dataset. GIST1M sampled at 9.3× work with the fixed ladder and at 15.2× with eigen-curve levels; the full-1M runs measured 13.1× and 20.8× respectively, so the sample is conservative. DEEP10M sampled MODERATE at 3.6× and measured 3.9× at 10^7 points. The ball level's sample SKIP was confirmed at full scale on all four datasets. On GIST, for example, 92% of points survive the ball level, the term counts are identical, and the wall-clock is strictly worse. In short, the diagnostic predicts full-scale outcomes from a sample. This includes the negative verdicts, which is its most valuable behavior: it prevents building a useless level.

Table 8: Cold-read exact 1-NN under `O_DIRECT` (every byte from the device; page cache bypassed, so each query is cold by construction). Scan streams the whole file; point fetches each needed vector from the original-order file; block fetches whole clusters from the cluster-major file. All arms return identical exact results.

Dataset	Arm	ms/query	MB/query	Reads
SIFT1M	scan	402	488	62
(488 MB)	point	**61** (7×)	**2.3** (212×)	588
	block	81 (5×)	37	73
GIST1M	scan	3006	3663	458
(3.6 GiB)	point	**421** (7×)	**91** (40×)	12,379
	block	751 (4×)	1224	158
DBpedia	scan	5385	5860	733
(5.7 GiB)	point	**318** (**17**×)	**8.0** (**730**×)	1028
	block	418 (13×)	374	36
DEEP10M	scan	3574	3664	458
(3.6 GiB)	point	1542 (2×)	**209** (18×)	50,356
	block	**1378** (3×)	780	223

6.6 Out of Core: Measured Cold Reads

Section 5.4 motivated the two-tier layout from structure. Table 8 now measures it end to end. The protocol is simple: all reads use `O_DIRECT` with page-aligned buffers. The page cache is bypassed, so every query pays device prices. This is a cold-cache protocol with no cache-flushing ceremony. It is also deliberately conservative on NVMe, where cheap random reads and high sequential bandwidth both favor the *scan* baseline.

There are three findings. **First, the headline.** On the OpenAI embeddings, exact cold-read 1-NN takes 318 ms against a 5.4-second scan, a 17× speedup. It reads 8 MB instead of 5.9 GB, which is 730× fewer bytes. The byte ratios are device-independent. On storage that is slower or more seek-bound than NVMe, they translate into even larger time ratios. **Second, the fetch set is highly cluster-local.** The embeddings' ~1,014 fetched points occupy a median of 27 of 1024 clusters, while uniform placement would touch about 644. SIFT's 583 points occupy 52 clusters. The reason is that the nearest neighbors of a query are near each other. This locality is what makes the cluster-major arm viable at all. It also means the k-means structure earns its place as a *partitioner* even where the funnel rejects its *bound* (SKIP on all four datasets). The same geometry is useless for compute but valuable for layout. **Third, point-versus-block is a device property.** With serial reads at 80 µs, the point arm's few thousand small fetches beat the block arm's fewer but heavier reads on three of the four datasets. Only DEEP10M's 50k-point fetch set flips the order. Parallel small reads would extend the point arm's advantage, since NVMe queue depths reward exactly that pattern. Seek-bound media would flip every row toward block. The layout writes both files, so the diagnostic should pick per device.

Table 9: ADSampling (faithful NumPy port of the reference implementation, ϵ_0=2.1, Δd=32) versus our PCA-only cascade, $k = 1$. Work in millions of dimension-terms per query; recall under float64-verified ground truth. Ours is exact by construction.

Dataset	ADS terms (recall)	Ours (exact)	Winner
SIFT1M	33.0 (1.000)	**12.6**	ours 2.6×
GIST1M	**44.5** (1.000)	46.1	tie
DBpedia-OpenAI	**49.8** (1.000)	119.6	ADS 2.4×
DEEP10M	325.8 (1.000)	**244.0**	ours 1.3×

One design lesson surfaced only here: *the optimal cascade depth is deeper out of core than in RAM.* SIFT's eigen-curve levels $(8, 16, 40)$ minimize in-RAM terms, but they fetch 3,381 points. The deeper $(8, 16, 32, 64)$ ladder spends more sketch terms to fetch only 588 points, and it wins the cold-read benchmark (61 vs. 91 ms). In RAM, terms are the currency. Out of core, fetched bytes are. The level-selection objective should follow the medium.

6.7 Baseline: ADSampling

ADSampling [5] is the natural adversary. It also prunes with incremental prefix distances after a rotation. However, its rotation is random, so partial distances become unbiased samples, and its stopping rule is a sequential hypothesis test with a tunable error rate ϵ_0. This buys harder pruning per dimension, at the price of the exactness guarantee. For a fair comparison, we ported the reference implementation into the same harness and the same currency: dimensions touched equal terms. Table 9 is honestly mixed in RAM. We win at 96–128 dimensions, ADS wins on the 1536-D embeddings, and GIST is a tie. At k=10, ADS widens its advantage. On recall, we must correct our own first measurement. An earlier float32 ground-truth check miscounted near-duplicate points as recall failures, due to catastrophic cancellation. Under float64 verification, ADS at the default ϵ_0 achieved perfect recall on every k=1 query we ran. It missed one true neighbor in 600 query-tasks (GIST, k=10, recall 0.9990). The distinction that matters is therefore *guarantee versus empirical*, not observed failure. The distinction is structural, because no ϵ_0 turns the test into a bound.

The decisive difference is not in RAM. ADS pays its $\Delta d = 32$ floor for *every* point in the dataset. SIFT's 33.0M terms and DEEP10M's 325.8M terms sit exactly at $32n$. ADS must therefore touch every vector, and out of core it reads full-scan bytes regardless of its parameters. Its successor PDX [9] restructures the layout dimension-major, but it still visits every vector. Our cascade fetches only 0.06–1% of the vectors (Table 8). Finally, the two methods compose rather than compete: a statistical test could accelerate the final full-distance checks over our survivor sets, at the usual price.

7 LIMITATIONS

High *intrinsic* dimensionality defeats every cheap bound. Distance concentration flattens the eigenvalue spectrum and the spread of d_p at the same time. The funnel exists precisely to detect this case and say so (Table 7, last row). DEEP10M sits near this boundary in RAM. At native-code constants, the in-RAM wall-clock race belongs to fused exhaustive scans (FAISS) at the 128-D scale. ADSampling's statistical pruning also beats our deterministic bounds in RAM on the highest-dimensional embeddings (§6.7). Exact k-NN loses the in-RAM wall-clock by $k = 100$ on every dataset we measured (§6.3). The out-of-core measurements use `O_DIRECT` on datasets that fit in RAM. This is valid device-price accounting, but a larger-than-RAM run, with parallel rather than serial fetch I/O, remains future work. Updates are also future work: the cluster-major layout assumes a static index that is rebuilt periodically, which is the same assumption disk-ANN builds make. Finally, ball-bound and layout quality depend on the k-means sample, as any learned structure does. Correctness never does.

8 CONCLUSION

Exact lower bounds are shapes, and shapes compose. The right composition is a property of the data, and it can be measured on a sample before any index is built. On that principle, this paper delivers an exact k-NN index that configures itself end to end. The cascade widths come from the eigenvalue curve; without this step, 1536-dimensional embeddings masquerade as unprunable. The levels come from a measured trial, and its sample verdicts are confirmed at 10^6–10^7 scale on four real datasets. The breaking points are stated honestly: $k = 100$ in RAM, and i.i.d. data everywhere. The out-of-core argument has become a measurement: exact cold-read queries run 7–17× faster than a disk scan and read 40–730× fewer bytes, against a state-of-the-art statistical pruner that structurally cannot leave RAM. The recurring lesson of this work is simple. Every constant we had silently tuned to one dataset—the cascade widths, the verdict thresholds, even our own ground-truth checker—failed on the next dataset. Every fix was the same: derive the constant from a measurement. Measure the data to choose the index; count the work to predict the time.

Artifact. Code, diagnostics, and all experiment scripts: `github.com/taiwuchiang-gmail/hierarchical-pca-search`

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